

# Operations on functions

Building new functions out of old ones, reading a graph, and the rules of logarithms.

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- 1 Adding and multiplying functions
  - 2 Reading a graph
  - 3 Composition of functions
  - 4 Inverses, and when one exists
  - 5 Finding an inverse
  - 6 The rules of logarithms
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# Adding and multiplying functions

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 **Practise: Quotients and sums of fractions · Add & subtract rational expressions** ([https://www.khanacademy.org/math/prec calculus/x9e81a4f98389efdf:rational-functions/x9e81a4f98389efdf:adding-and-subtracting-rational-expressions/e/adding\\_and\\_subtracting\\_rational\\_expressions\\_5](https://www.khanacademy.org/math/prec calculus/x9e81a4f98389efdf:rational-functions/x9e81a4f98389efdf:adding-and-subtracting-rational-expressions/e/adding_and_subtracting_rational_expressions_5))

 **Practise: Products of functions · Multiply binomials** (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/e/multiply-binomials-coefficient>)

Khan has no lesson on  $f + g$ ,  $f - g$  and  $fg$  as such. These two are the same algebra on expressions, which is the closest it gets.

## THE FIVE

$$(f + g)(x) = f(x) + g(x) \quad (fg)(x) = f(x)g(x)$$

and the same idea for the difference, the quotient, and a constant times a function.

- 1 **Do the operation one value at a time.** Whatever  $x$  is, work out  $f(x)$  and  $g(x)$ , then combine those two numbers.
- 2 **The result is defined only where both are.** Intersect the two domains.
- 3 **A quotient also needs the bottom to be non-zero.**  $(f/g)(x)$  needs  $g(x) \neq 0$ .
- 4 **The combination you will use constantly is profit.**

$$P(q) = R(q) - C(q)$$

- 5 **Cost usually splits into fixed plus variable.** The fixed part is what you pay at  $q = 0$ , so it is the  $y$ -intercept.

## The Rivella Coffee case

### The five operations

$$(f + g)(x) = f(x) + g(x) \quad (f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x) \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad (cf)(x) = c \cdot f(x)$$

WORKED, WITH  $F(X) = 3X - 1$  AND  $G(X) = X^2 + 3X$

$$(f + g)(x) = x^2 + 6x - 1$$

$3x - 1 + x^2 + 3x$ , collected.

$$(f - g)(x) = -x^2 - 1$$

$3x - 1 - x^2 - 3x$ . The  $3x$  terms cancel.

$$(fg)(x) = 3x^3 + 8x^2 - 3x$$

$(3x - 1)(x^2 + 3x)$ , expanded.

$$\left(\frac{f}{g}\right)(x) = \frac{3x - 1}{x^2 + 3x}$$

Undefined where  $x^2 + 3x = 0$ , so at 0 and  $-3$ .

$$\left(\frac{1}{2}f\right)(x) = \frac{3}{2}x - \frac{1}{2}$$

A constant multiplies every term.

### Why it matters: profit

$$P(q) = R(q) - C(q)$$

#### WATCH OUT

- Put the second function in brackets before subtracting, or the signs of its later terms survive when they should not.
- A quotient is undefined wherever the bottom function is zero, even where the top is perfectly fine.
- The combination is only defined where both originals are. Intersect the domains.

# Reading a graph

KHAN ACADEMY

► **Watch:** Both intercepts · x-intercepts and y-intercepts (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/v/introduction-to-intercepts>)

► **Watch:** Intercepts of a parabola · Interpret a quadratic graph (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:intro-parabolas/v/interpret-a-quadratic-graph>)

≡ **Read:** Intercepts of lines review (x-intercepts and y-intercepts) (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/a/intercepts-of-lines-review-x-intercepts-and-y-intercepts>)

✍ **Practise:** From the formula · Intercepts from an equation ([https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/solving\\_for\\_the\\_x-intercept](https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/solving_for_the_x-intercept))

✍ **Practise:** Intercepts from a graph (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/linear-function-intercepts>)

- 1 **The y-intercept is the value at zero.** Put  $x = 0$ . There is at most one.
- 2 **The x-intercepts are where the function is zero.** Solve  $f(x) = 0$ . There may be several, or none.
- 3 **Plot the intercepts first, then enough points to see the shape.**
- 4 **In a business question the intercepts mean something.** The y-intercept is usually a fixed cost. An x-intercept is usually break-even.
- 5 **A vertical line is not a function.**  $x = 3$  is a perfectly good line and has no y to be a function of x.

## The method

**WORKED:**  $Y = 2X + 3$

$$y = 0 \rightarrow 0 = 2x + 3$$

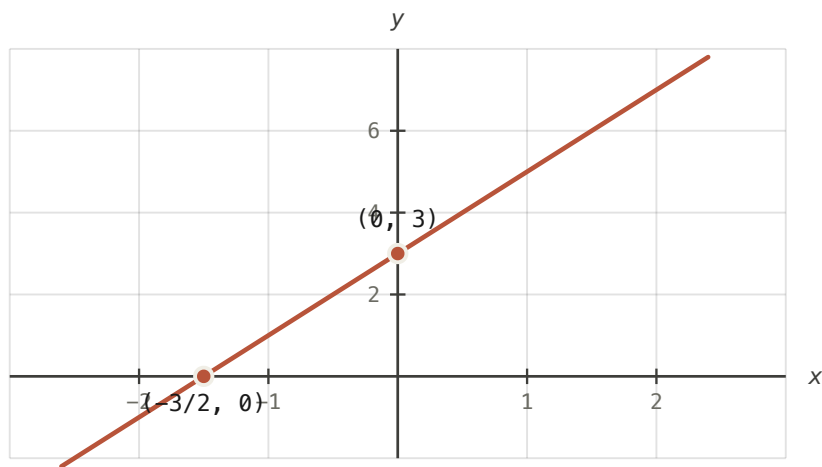
Set  $y$  to zero for the  $x$ -intercept.

$$x = -\frac{3}{2}$$

One  $x$ -intercept.

$$x = 0 \rightarrow y = 2(0) + 3 = 3$$

Set  $x$  to zero for the  $y$ -intercept.



$y = 2x + 3$ . It crosses the vertical axis at 3 and the horizontal axis at  $-3/2$ .

**A SECOND:**  $x = 3$

$$x = 3$$

A vertical line.

**no  $y$ -intercept**

Because  $x$  can never be 0 on this graph.

## Break-even is an $x$ -intercept

### WATCH OUT

- A zero denominator is not an  $x$ -intercept. It is a break in the curve.
- For a quadratic, a negative discriminant means no  $x$ -intercept at all, and that is a complete answer.
- Only the numerator can make a fraction zero, so solve the numerator and then check the value is allowed.

# Composition of functions

KHAN ACADEMY

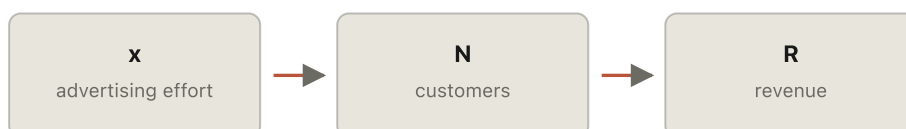
- ▶ **Watch:** What a composite is · Composing functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/v/function-composition>)
- ▶ **Watch:** At a number · Evaluating composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/v/evaluating-composite-functions>)
- ≡ **Read:** The definition · Intro to composing functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/a/introduction-to-function-composition>)
- ✍ **Practise:** At a number · Evaluate composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/e/evaluate-composite-functions-from-formulas>)
- ✍ **Practise:** Building the formula · Find composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/e/compose-functions>)

## COMPOSITION

$$(g \circ f)(x) = g(f(x))$$

Do  $f$  first, then feed its output straight into  $g$ . The circle is read right to left.

- 1 **Work from the inside out.** Compute the inner value, then put it into the outer function.
- 2 **Substitute the whole inner output.** Every  $x$  in the outer function becomes the entire inner expression, brackets included.
- 3 **Order matters.**  $f \circ g$  is almost never  $g \circ f$ . Check with one number if you doubt it.
- 4 **For it to exist, the middle value has to be allowed.**  $f(x)$  must be something  $g$  accepts.
- 5 **The quantity in the middle is real, but the question is not about it.** Advertising gives customers, customers give revenue; the question asks revenue from advertising.



The chain. The middle quantity is real, but the question is not about it.

#### THE CALCULATION

$$R(N(x)) = 70(100\sqrt{x}) + \frac{(100\sqrt{x})^2}{20}$$

Every  $n$  becomes  $100\sqrt{x}$ , including the one being squared.

$$= 7000\sqrt{x} + 500x$$

Because  $(100\sqrt{x})^2 = 10\,000x$  and  $\frac{10\,000x}{20} = 500x$ .

#### The definition

$$(g \circ f)(x) = g(f(x))$$

#### Two properties

ORDER MATTERS:  $F(x) = 2x + 1$ ,  $G(x) = x^2 - 3$

$$(g \circ f)(x) = (2x + 1)^2 - 3 = 4x^2 + 4x - 2$$

Write the outer function with an empty bracket, then fill it.

$$(f \circ g)(x) = 2(x^2 - 3) + 1 = 2x^2 - 5$$

The other order.

**at  $x = 4$ : 78 against 27**

Same functions, same input, different answers.

#### WATCH OUT

- $(g \circ f)(x)$  is not  $g(x) \cdot f(x)$ . The ring is not a multiplication dot.
- The whole inner function goes inside the bracket:  $(2p + 1)^2$ , never  $2p^2 + 1$ .
- Take the domain from the two original functions, not from the tidied-up composite.

# Inverses, and when one exists

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- ▶ **Watch:** What an inverse is · Intro to inverse functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/v/introduction-to-function-inverses>)
- ▶ **Watch:** When one exists · Invertible functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/v/determining-if-a-function-is-invertible>)
- ≡ **Read:** One-to-one, written out · Intro to invertible functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/a/intro-to-invertible-functions>)
- ✍ **Practise:** Spotting when there is no inverse · Determine if a function is invertible (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/e/inverse-domain-range>)
- ✍ **Practise:** Reading an inverse off a graph · Inverse functions: graphs and tables (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:inverse-functions-in-graphs-and-tables/e/inverse-functions-graphs-and-tables>)

## INVERSE

The function that sends each output back to the input it came from.

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

## ONE-TO-ONE

Different inputs always give different outputs. No output is ever produced twice.

- 1 **A function has an inverse exactly when it is one-to-one.** That is the whole condition.
- 2 **On a graph, the horizontal line test decides.** A horizontal line may meet the graph at most once. Twice means two inputs share an output, so there is no way back.
- 3 **Strictly increasing or strictly decreasing is enough.** A function that only ever rises, or only ever falls, is one-to-one.
- 4 **The domain and the range swap.** The inverse's inputs are the original's outputs.
- 5 **The graph of the inverse is the reflection in the line  $y = x$ .**

WATCH OUT

- An inverse is not a reciprocal.  $f^{-1}(x)$  is not  $\frac{1}{f(x)}$ .
- A function that repeats an output has no inverse, because there is no way back to which input it came from.
- Swapping the domain and the range is part of the answer, not an afterthought.

$$Q(p) = \frac{50\,000\,000}{p+1} - 5\,000\,000 \quad \text{for } 0 \leq p \leq 9$$

#### THE DERIVATION

$$q = \frac{50\,000\,000}{p+1} - 5\,000\,000$$

Start from the definition.

$$q + 5\,000\,000 = \frac{50\,000\,000}{p+1}$$

Move the constant across.

$$p+1 = \frac{50\,000\,000}{q+5\,000\,000}$$

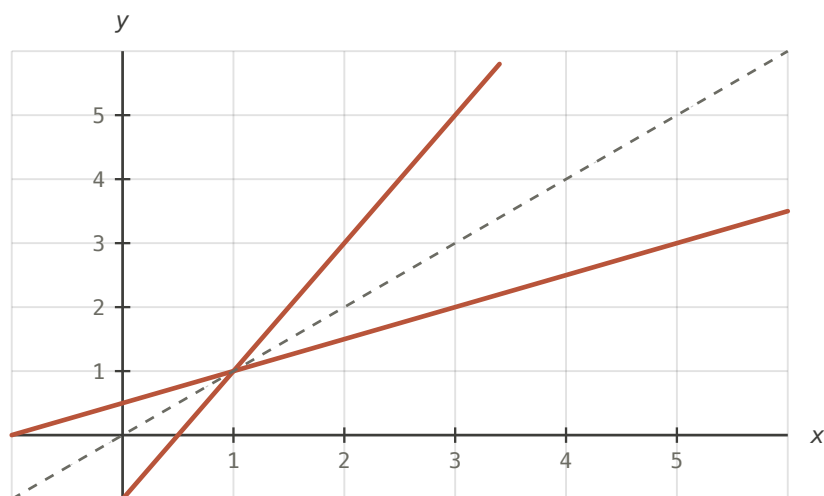
Take reciprocals of both sides.

$$P(q) = \frac{50\,000\,000}{q+5\,000\,000} - 1$$

The price as a function of the quantity.

#### The definition and the check

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$



$f(x) = 2x - 1$  and its inverse, mirrored in the dashed line  $y = x$ .

# Finding an inverse

KHAN ACADEMY

- ▶ **Watch:** Finding one, step by step · Finding inverse functions: linear (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/v/function-inverse-example-1>)
- ▶ **Watch:** Checking it · Verifying inverse functions by composition (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/v/verify-inverse-functions-from-tables>)
- ≡ **Read:** The method, written out · Intro to inverse functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/a/intro-to-inverse-functions>)
- ✍ **Practise:** Finding one · Find inverses of linear functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/e/algebraically-finding-inverses>)
- ✍ **Practise:** Checking by composition · Verify inverse functions ([https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/e/inverses\\_of\\_functions](https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/e/inverses_of_functions))

- 1 Write  $y = f(x)$ .
- 2 Solve for  $x$  in terms of  $y$ . This is the only step that is real work.
- 3 Swap the names.  $x$  goes back to being the input, so the answer is written as a function of  $x$ .
- 4 Check by composing both ways. You should get  $x$  back each time. This catches almost every slip.
- 5 If it is not one-to-one, restrict the domain until it is.  $f(x) = (x - 1)^2$  has no inverse until you keep only  $x \geq 1$ .
- 6 The linear case is worth knowing by sight.

$$f(x) = ax + b \quad \Rightarrow \quad f^{-1}(x) = \frac{x - b}{a}$$

- 7 To solve  $f(x) = k$ , apply the inverse to both sides. That is what an inverse is for.

## Finding one, and when it exists

THE GENERAL LINEAR CASE:  $F(x) = Ax + B$

$$g(x) = \frac{x - b}{a}$$

Undo in reverse order: subtract b, then divide by a.

$$f(g(x)) = a \frac{x - b}{a} + b = x$$

Check one way.

$$g(f(x)) = \frac{(ax + b) - b}{a} = x$$

And the other. So it is the inverse, and it is linear too.

## Using an inverse to solve an equation

### WATCH OUT

- $f^{-1}(x)$  is the inverse function, not  $\frac{1}{f(x)}$ .
- A function that fails the horizontal line test has no inverse until its domain is restricted.
- Domain and range swap. If you are asked for the domain of the inverse, look at the range of the original.

# The rules of logarithms

KHAN ACADEMY

- ▶ **Watch:** The logarithm rules · Intro to logarithm properties (1 of 2) (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/v/introduction-to-logarithm-properties>)
- ▶ **Watch:** The rule that solves equations · Using the logarithmic power rule (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/v/logarithm-of-a-power>)
- ▶ **Watch:** The change of base formula for logarithms (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:change-of-base/v/change-of-base-formula>)
- ≡ **Read:** The rules, written out · Intro to logarithm properties (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/a/properties-of-logarithms>)
- ✍ **Practise:** Expanding and condensing · Use the properties of logarithms ([https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/e/logarithms\\_2](https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/e/logarithms_2))
- ✍ **Practise:** Evaluate logarithms: change of base rule (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:change-of-base/e/evaluate-logarithms-using-the-change-of-base-rule>)

## THE THREE THAT EXIST

$$\log(ab) = \log a + \log b \quad \log \frac{a}{b} = \log a - \log b \quad \log(a^n) = n \log a$$

## THE ONE THAT DOES NOT

There is no rule for  $\log(a + b)$ . It does not simplify. Nothing you do turns a sum inside a logarithm into anything shorter.

- 1 **Expanding:** a product becomes a sum, a quotient a difference, a power a coefficient.
- 2 **Simplifying:** the same three rules read right to left.
- 3 **The power rule is the one that solves equations.** It brings the unknown down out of the exponent where you can reach it.
- 4 **Change the base when the calculator has not got it.**

$$\log_b x = \frac{\ln x}{\ln b}$$

- 5 The exponential and the logarithm undo each other.

$$\exp(\ln u) = u \quad a^x = \exp(x \ln a) \quad \text{for } a > 0$$

- 6 Check the argument is positive before applying any of it.

| Exponents                                      |                             |
|------------------------------------------------|-----------------------------|
| RULE                                           | RULE                        |
| $b^x \cdot b^y = b^{x+y}$                      | $\frac{b^x}{b^y} = b^{x-y}$ |
| $(b^x)^y = b^{xy}$                             | $(bc)^x = b^x c^x$          |
| $\left(\frac{b}{c}\right)^x = \frac{b^x}{c^x}$ | $b^1 = b$                   |
| $b^0 = 1$                                      | $b^{-x} = \frac{1}{b^x}$    |

| Logarithms                             |                                            |
|----------------------------------------|--------------------------------------------|
| RULE                                   | RULE                                       |
| $\log_b(mn) = \log_b m + \log_b n$     | $\log_b \frac{m}{n} = \log_b m - \log_b n$ |
| $\log_b(m^r) = r \log_b m$             | $\log_b \frac{1}{m} = -\log_b m$           |
| $\log_b 1 = 0$                         | $\log_b b = 1$                             |
| $\log_b m = \frac{\log_a m}{\log_a b}$ | change of base                             |

## Expanding

### WRITING LOGARITHMS IN TERMS OF SIMPLER LOGARITHMS

$$\ln \frac{x}{zw} = \ln x - \ln(zw)$$

Quotient rule first.

$$= \ln x - \ln z - \ln w$$

Then the product rule, and the minus reaches both.

$$\ln \sqrt{x^5(x-2)^8 \div (x-3)^3} \rightarrow \frac{1}{3} [5 \ln x + 8 \ln(x-2) - \ln(x-3)]$$

A cube root is a power of one third, so the whole bracket is scaled by it.

## Simplifying

### WORKED EXAMPLES

$$\ln e^{3x} = 3x$$

The two undo each other.

$$\log 1 + \log 1000 = 0 + 3 = 3$$

$$1000 = 10^3.$$

$$\log_7 {}^9\sqrt{7^8} = \frac{8}{9}$$

$$\text{Because } {}^9\sqrt{7^8} = 7^{8/9}.$$

$$\log_3 \frac{27}{81} = \log_3 \frac{1}{3} = -1$$

$$\frac{3^3}{3^4} = 3^{-1}.$$

$$\ln e + \log \frac{1}{10} = 1 + (-1) = 0$$

Both special values.

### In summary

$$\exp(\ln u) = u \quad \ln(\exp x) = x \quad \exp(x) = u \Leftrightarrow x = \ln u$$

$$a^x = \exp(x \ln a) \text{ for } a > 0$$

### WATCH OUT

- $\log(a + b)$  does not break up. There is no rule for the logarithm of a sum.
- $\frac{\log a}{\log b}$  is not  $\log(a - b)$ , and  $(\log a)^2$  is not  $2 \log a$ .
- Isolate the exponential before taking logarithms, and reject any answer that makes an original logarithm take a non-positive input.

## Week 2 review quiz

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Six questions in the Cyberlearn format. The operations, composition and inverses from this week, with the distractors the usual mistakes produce.

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**Quantitative Methods I.** Notes I wrote while learning this. Free to use and to share.

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